Henry Chen

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github.com/Chenry513 | chenry513.github.io/HenryChen

Education

Master of Science in Applied Computing (CGPA: 91%) — Sept. 2025 – Dec. 2026 (est.)
British Columbia Institute of Technology — Burnaby, BC

Bachelor of Science in Computing Science — Sept. 2020 – Aug. 2025
Simon Fraser University — Burnaby, BC

Technical Skills

Languages: Python, JavaScript, TypeScript, C/C++, SQL, Bash
Web & Frameworks: Django, Django REST Framework, Vue.js, Next.js, Flask, REST APIs
Machine Learning & Data: scikit-learn, pandas, numpy, Gradient Boosting, ETL pipelines, cross-validation, feature engineering
Databases & DevOps: PostgreSQL, MySQL, SQLite, Git, GitHub Actions, CI/CD, Linux, Vercel, Render
Testing & Quality: Google Test (C++), unit testing, schema validation, code coverage

Work Experience

Mitacs Research Intern – Robotics & Computer Vision — Jan. 2026 – Present
BCIT — Seaspan Shipyards Collaboration (Supervisor: Mathew Smith) — Vancouver, BC

- Developing a 3D weld joint reconstruction pipeline using VGGT (Meta Research) to extract geometric coordinates for robotic arm path planning.
- Built WeldPath Viz, a browser-based 3D point cloud tool for inspecting reconstructed joints and defining weld seam start/end coordinates for automated welding. chenry513.github.io/weldpath-viz/visualize_weld.html
- Designed a calibration system to translate reconstructed scene coordinates into real-world millimeter measurements for robotic execution.

Research Assistant (Backend & Automation Engineer) — May – Aug. 2025
Simon Fraser University — PadComputing Lab (Prof. Nicholas Vincent) dl-tools.onrender.com — Burnaby, BC

- Developed an automated ingestion pipeline validating structured user submissions against defined requirements prior to production integration, reducing manual review effort by 60%.
- Built a Python-based schema validation tool to enforce expected data behavior and prevent invalid content from entering system workflows.
- Integrated GitHub Actions CI to execute automated validation checks on each submission, supporting consistent system outcomes.
- Maintained and extended an existing Django codebase while preserving system stability and documenting processes for non-technical contributors.

Research Assistant (Backend Engineer) — Aug. – Dec. 2024
Simon Fraser University — PadComputing Lab (Prof. Nicholas Vincent) — Burnaby, BC

- Developed backend components for a Django-based research platform supporting structured LLM metadata and user-facing functionality.
- Designed relational models and schema logic to support consistent data behavior and maintainable structured storage.
- Implemented 8+ RESTful API endpoints using Django REST Framework, translating functional requirements into reliable system interactions.
- Managed migrations and enhanced Django Admin tooling to maintain data integrity and support stable platform evolution.

PROJECTS

Full-Stack Personal Project — Feb. 2026
readmeify — AI-Powered README Generator — readmeify-five.vercel.app

- Built and deployed a full-stack web app that produces tailored README documentation by reading actual repo structure, dependencies, and file contents — eliminating manual documentation overhead.
- Implemented GitHub OAuth for authentication and the GitHub API to extract file trees and dependency manifests as structured context, enabling accurate and project-specific generation.
- Integrated Groq API (LLaMA 3.3 70B) with real-time streaming output and built with Next.js, NextAuth, and deployed on Vercel with secure server-side environment variable handling.

Data Science & ML Personal Project — Dec. 2025
CardioScan — Cardiac Risk Assessment Dashboard — cardiac-assessment.onrender.com

- Designed an ETL pipeline extracting and combining 4 UCI data sources, transforming inconsistent encodings across institutions, and imputing up to 66% missing values to produce a clean 920-patient cohort.
- Trained a Gradient Boosting Classifier (XGBoost-equivalent) achieving AUC 0.892 and 84.2% accuracy with 5-fold stratified cross-validation, outperforming Random Forest and Logistic Regression baselines.
- Deployed a clinical Flask dashboard with PostgreSQL persistence, SHAP-style per-patient feature contribution analysis, and population-level analytics across 4 institutional data sources.

Real-Time Web Application Personal Project — July 2024
Rankr — Collaborative Real-Time Ranking App — github.com/Chenry513/rankr

- Built a real-time polling system enabling groups to create, join, and rank shared options collaboratively.
- Implemented live updates using React, TypeScript, Socket.io, and Redis for low-latency synchronization.
- Designed backend logic to aggregate ranked votes and compute final group decisions.